



Module 01

INTRODUCTION TO MACHINE LEARNING

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Lab Objectives

By the end of his lab, students should be able to understand the basic concept of Machine Learning, data analytic and the basic programming of machine learning in Orange Data Mining.

In this lab, you are required to have ample understanding on Machine Learning. The objectives of this lab exercises is to ensure students familiarise with Machine Learning algorithms and Orange Data Mining software. This lab sheet is structured as follows:

- i. Introduction to Machine Learning and Orange Data Mining
- ii. PART A: Installation and Making the Orange workflow
- iii. PART B: Building Machine Learning Models
- iv. PART C: Improving the trained model
- v. References

Introduction to Machine Learning and Orange Data Mining

Machine learning is an application that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.

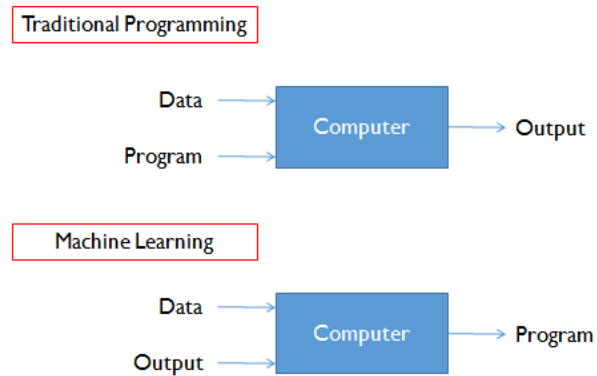


Figure 1 Computer Program Architecture

Orange is a component-based visual programming software package for data visualization, machine learning, data mining, and data analysis. Orange components are called widgets and they range from simple data visualization, subset selection, and preprocessing, to empirical evaluation of learning algorithms and predictive modeling.



Figure 2 Orange Data Mining

PART A: Installation and Making the Orange Workflow

(a) Setting up the application

In this session, we're going to use Orange Data Mining software as a machine learning tool. If you haven't installed the software yet, please follow these steps to download the Orange Data Mining.

Step 1: Go to <https://orange.biolab.si/> and click on download to get the latest version.

Step 2: Install the software and setup the workspace.

(b) Making the workflow

The next step, we're going to make our first workflow towards building a solution to the problem. Firstly, understand what steps we need to take in order to achieve our goal. Click on "New" and this is what you should have come up with.

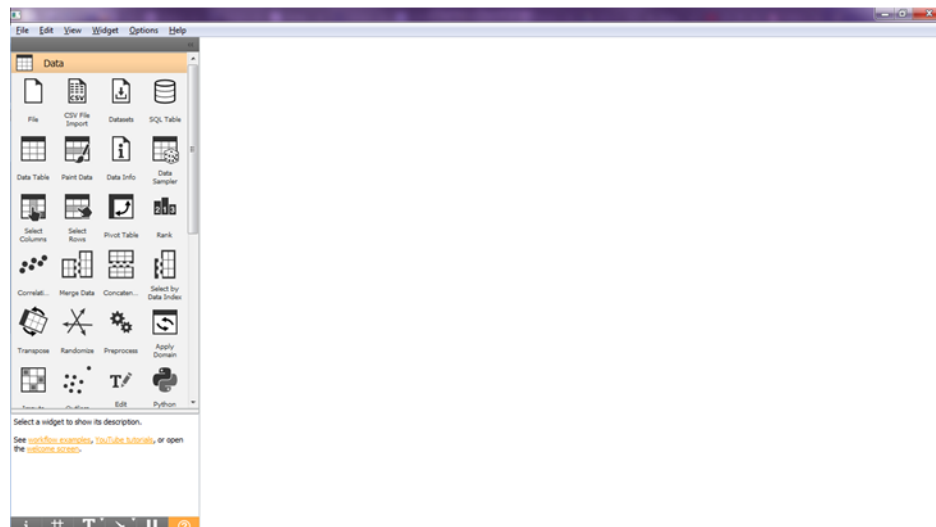


Figure 3 New blank project

Step 1: Add a **File** widget and click on it.



Figure 4 File widget

Step 2: Choose any of 2 options given. Click on 'File' and choose from your computer or click on 'URL' to directly insert the source from the internet. For this problem, we're using Car Evaluation datasets.

Step 3: A table is shown below after the file was chosen. Then, click on 'Role' and assign one variable as 'Target'. Click apply.

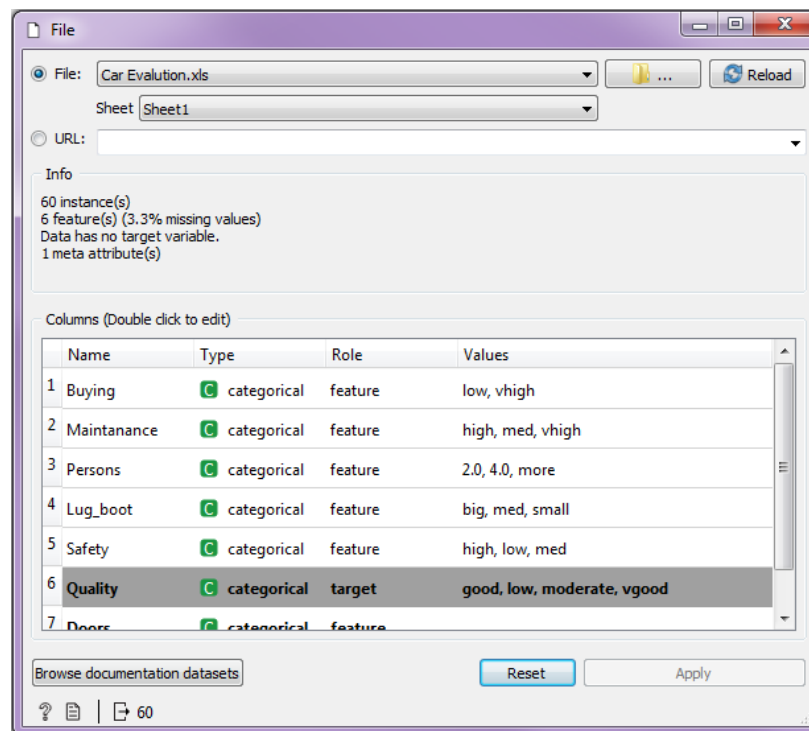


Figure 5 File explorer

Step 4: As we need a data table to better visualize our findings, we add a **Data Table** widget and double click the widget to visualize your table.

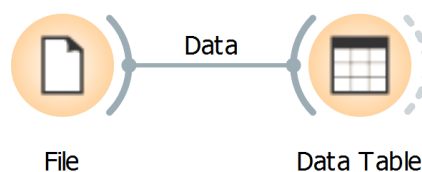


Figure 6 Connection File widget to Data Table

	Quality	Buying	Maintenance	Persons	Lug_boot	Safety	Doors
1	moderate	vhigh	med	4.0	small	high	2.0
2	moderate	vhigh	med	4.0	med	high	2.0
3	moderate	low	vhigh	more	med	high	2.0
4	moderate	low	vhigh	more	?	med	2.0
5	moderate	low	vhigh	more	big	high	2.0
6	moderate	low	vhigh	4.0	small	high	3.0
7	moderate	low	vhigh	4.0	med	high	3.0
8	moderate	?	vhigh	4.0	big	med	3.0
9	moderate	low	vhigh	4.0	big	high	3.0
10	moderate	low	vhigh	4.0	small	high	3.0
11	moderate	low	vhigh	more	med	med	3.0
12	moderate	low	vhigh	more	med	high	3.0
13	moderate	low	vhigh	more	big	med	3.0
14	moderate	low	?	4.0	small	med	4.0
15	moderate	low	med	more	small	med	4.0
16	good	vhigh	vhigh	2.0	med	low	2.0
17	good	vhigh	vhigh	4.0	med	low	2.0
18	good	vhigh	vhigh	4.0	med	med	2.0
19	good	vhigh	vhigh	more	med	high	2.0
20	good	vhigh	vhigh	2.0	small	med	3.0
21	good	vhigh	vhigh	2.0	big	med	3.0
22	good	vhigh	vhigh	more	med	med	3.0
23	good	vhigh	vhigh	more	med	high	3.0
24	good	vhigh	vhigh	2.0	med	low	4.0
25	good	vhigh	vhigh	2.0	med	high	4.0
26	good	low	med	4.0	?	high	2.0
27	good	low	med	4.0	med	high	2.0
28	good	low	med	4.0	big	med	2.0
29	good	low	med	4.0	big	med	4.0
30	good	low	?	more	small	high	4.0
31	low	low	med	4.0	med	low	3.0
32	low	low	med	4.0	big	low	3.0
33	low	low	med	more	small	low	3.0
34	low	low	med	more	med	low	3.0
35	low	low	med	more	big	low	3.0
36	low	low	med	2.0	small	low	4.0

Figure 7 Data visualization on Data Table widget

As we can see in Figure 7, there are some missing values (circle in red). In order to solve the problem, we have to impute the missing values. This will be explained in the next step.

(c) Data cleaning or Data Pre-processing

Step 1: Add a *Preprocess* widget and double click on it.

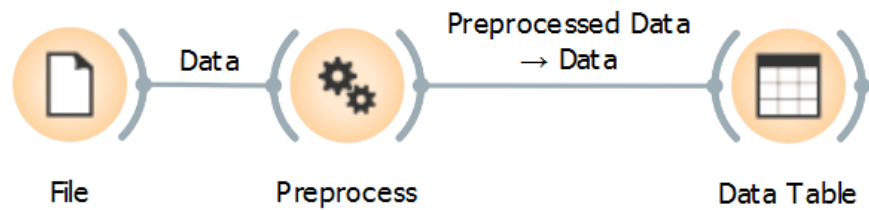


Figure 8 Preprocess widget

Step 2: When you double click on the widget after placing it, you will see that there are a variety of preprocessing methods you can use. Then, click on 'Impute Missing Values'

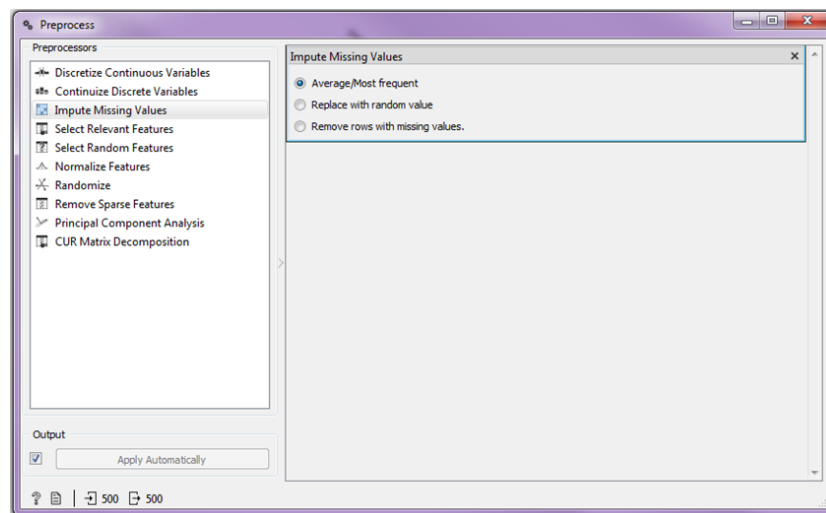


Figure 9 Selecting the preprocessors

Here, we have selected the default method to be Average for numerical values and Most Frequent for text based values (categorical).

You can select from other imputations like:

- Replace with random Values
- Remove the rows with missing values

	Quality	Buying	Maintenance	Persons	Lug_boot	Safety	Doors
1	moderate	vhhigh	med	4.0	small	high	2.0
2	moderate	vhhigh	med	4.0	med	high	2.0
3	moderate	low	vhhigh	more	med	high	2.0
4	moderate	low	vhhigh	more	med	med	2.0
5	moderate	low	vhhigh	more	big	high	2.0
6	moderate	low	vhhigh	4.0	small	high	3.0
7	moderate	low	vhhigh	4.0	med	high	3.0
8	moderate	low	vhhigh	4.0	big	med	3.0
9	moderate	low	vhhigh	4.0	big	high	3.0
10	moderate	low	vhhigh	more	small	high	3.0
11	moderate	low	vhhigh	4.0	med	med	3.0
12	moderate	low	vhhigh	more	med	high	3.0
13	moderate	low	vhhigh	more	big	med	3.0
14	moderate	low	med	4.0	small	med	4.0
15	moderate	low	med	more	small	med	4.0
16	good	vhhigh	vhhigh	2.0	med	low	2.0
17	good	vhhigh	vhhigh	4.0	med	low	2.0
18	good	vhhigh	vhhigh	4.0	med	med	2.0
19	good	vhhigh	vhhigh	more	med	high	2.0
20	good	vhhigh	vhhigh	2.0	small	med	3.0
21	good	vhhigh	vhhigh	2.0	big	med	3.0
22	good	vhhigh	vhhigh	more	med	med	3.0
23	good	vhhigh	vhhigh	more	med	high	3.0
24	good	vhhigh	vhhigh	2.0	med	low	4.0
25	good	vhhigh	vhhigh	2.0	med	high	4.0
26	good	low	med	4.0	med	high	2.0
27	good	low	med	4.0	med	high	2.0
28	good	low	med	4.0	big	med	2.0
29	good	low	med	4.0	big	med	4.0
30	good	low	med	more	small	high	4.0
31	low	low	med	4.0	med	low	3.0
32	low	low	med	4.0	big	low	3.0
33	low	low	med	more	small	low	3.0
34	low	low	med	more	med	low	3.0
35	low	low	med	more	big	low	3.0
36	low	low	med	2.0	small	low	4.0

Figure 10 Data Table after imputation

After imputation, the dataset have no missing values. All the missing values have been imputed by average/most frequent method through **Preprocess** widget.

Question 1 (10 marks)

You're required to create your own dataset with minimum 4 features and 1 target class. The features can be anything (age, gender, cloth size, name, height and etc.). Also try to add a blank data as shown in Figure 7.

Learning Objectives:

- Create a .csv or .xls file and read it with a **File** widget.
- Examine a dataset and assign one feature as '*Target*' role.
- Impute missing values average/ most frequent method.

PART B: Building Machine Learning Models

Important Note: You are required to complete PART A before proceeding to Part B

Beginning with the basics and well-known model, we will first train a *logistic regression* model encompassing all the features just to understand how to select and build models.

Step 1: First, add a **Data Sampler** widget and connect it to **Data Table** widget.

Step 2: Add a **Logistic Regression** widget and **Test and Score** widget. Then, both widgets are connected to the **Data Sampler** widget. Details of widgets connection is demonstrated below:

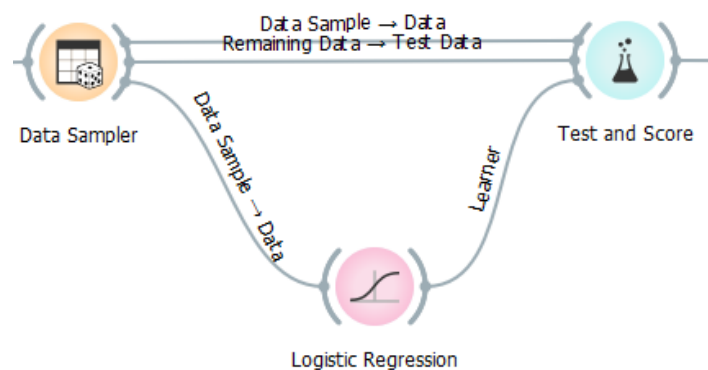


Figure 11 Workflow of building a model

Step 3: Go to **Data Sampler** widget and double click on it. This widget is used to split the dataset into training set and test set.

Step 4: Choose the sampling type that we want. In this context, we set the sampling type to 'Fixed proportion of data' with 60%. The percentage indicates that 60% of data are used to train the model and the remaining data (40%) are use to test the model.

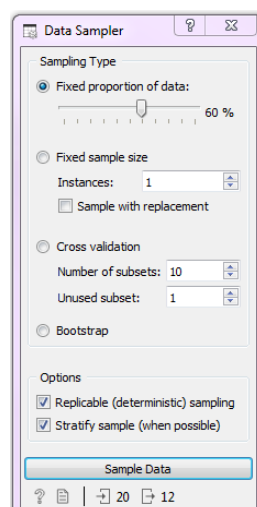


Figure 12 Data Sampler

Step 5: Double click on **Logistic Regression** widget. Few parameters are shown and these parameters can be manipulated in order to obtain good results. But for basic and easy understanding, we encouraged that do not change any of the setting. Let the model in default setting.

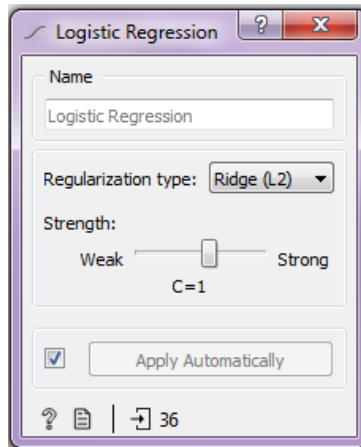


Figure 13 Logistic Regression

Step 6: Double click on **Test and Score** widget. For training the model, first we select the 'Leave one out' option (red box). The selection of this method is because of the dataset is small. The score of the trained model was illustrated (orange box).

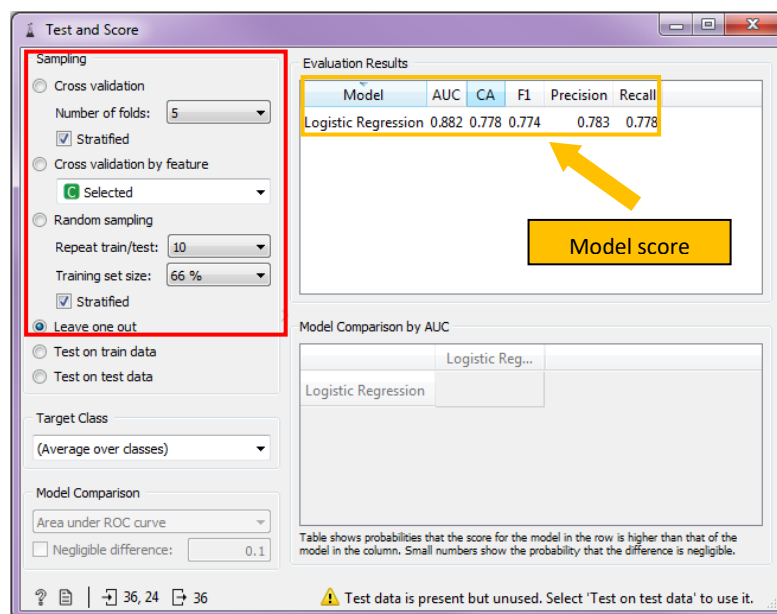


Figure 14 Score of trained model

To validate the trained model, click on 'Test on test data' option (red box). The final results of the model are shown (orange box).

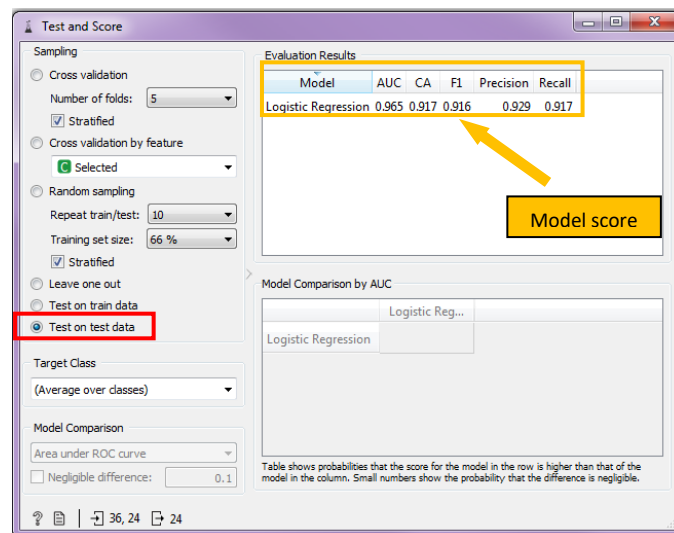


Figure 15 Test score on trained model

Step 7: Add a **Confusion Matrix** widget and connect it to **Test and Score** widget. The use of **Confusion Matrix** widget is to visualize the results better.

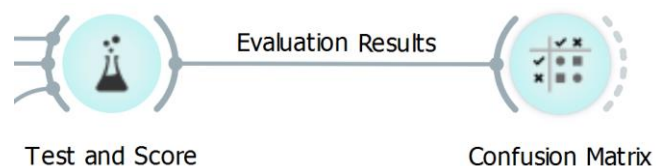


Figure 16 Test and Score linked with Confusion Matrix

Step 8: Double click on **Confusion Matrix** widget. As we can see, in training and testing the model, there are few classes that misclassified (highlighted in red) and data that correctly classified highlighted in blue.

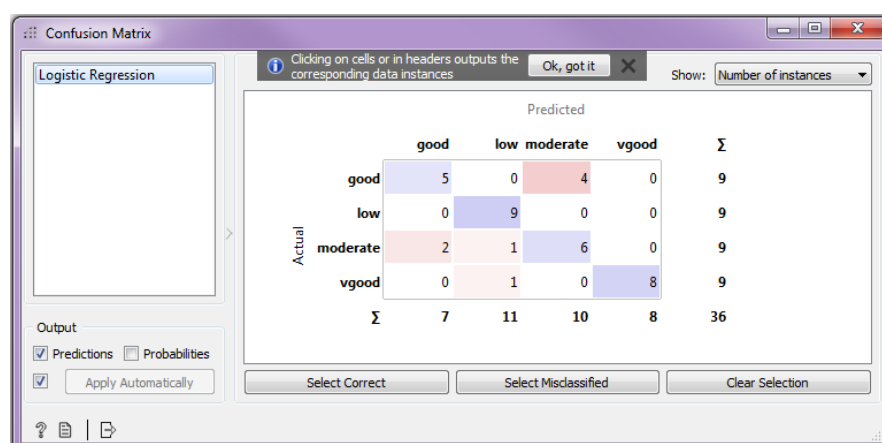


Figure 17 Confusion Matrix of trained Logistic Regression

For testing the model, only two data were misclassified.

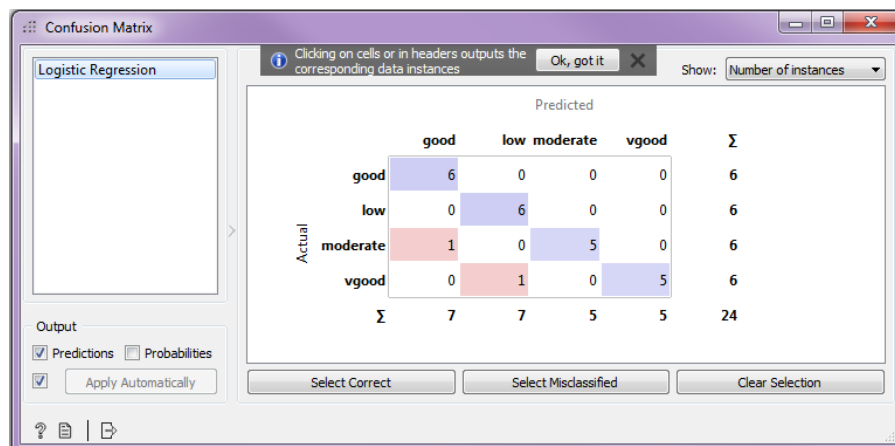


Figure 18 Confusion Matrix of testing Logistic Regression

This way, you can test out different models and see how accurately they perform.

Let's try to evaluate, how a Support Vector Machine, Neural Network, and k-Nearest Neighbors would do? Change the modeling method to Support Vector Machine, Neural Network, and k-Nearest Neighbors and look at the confusion matrix. Figure 19 and 20 illustrated the train and test results for multiple models.

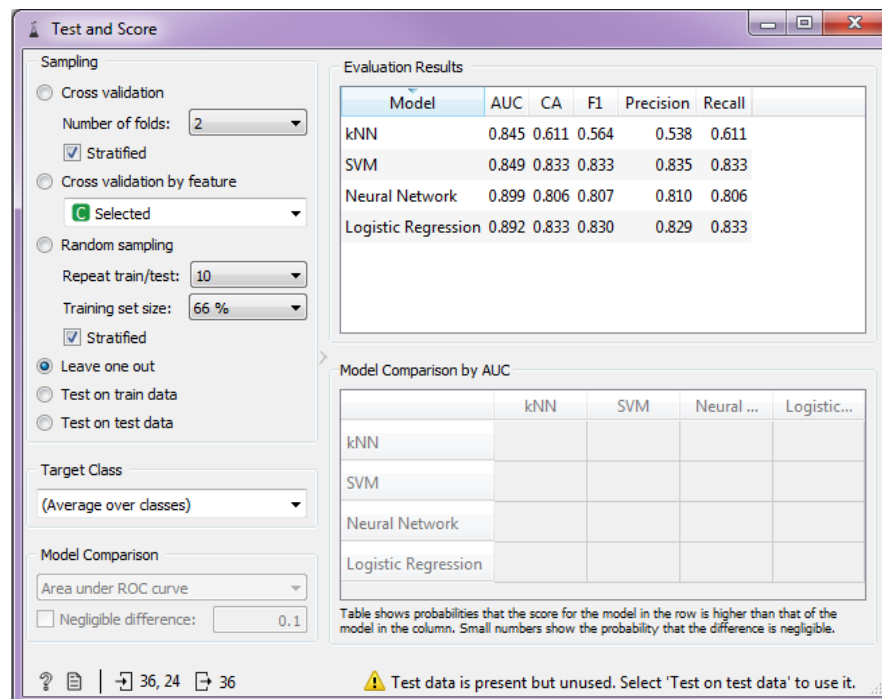


Figure 19 Training results for four classifiers

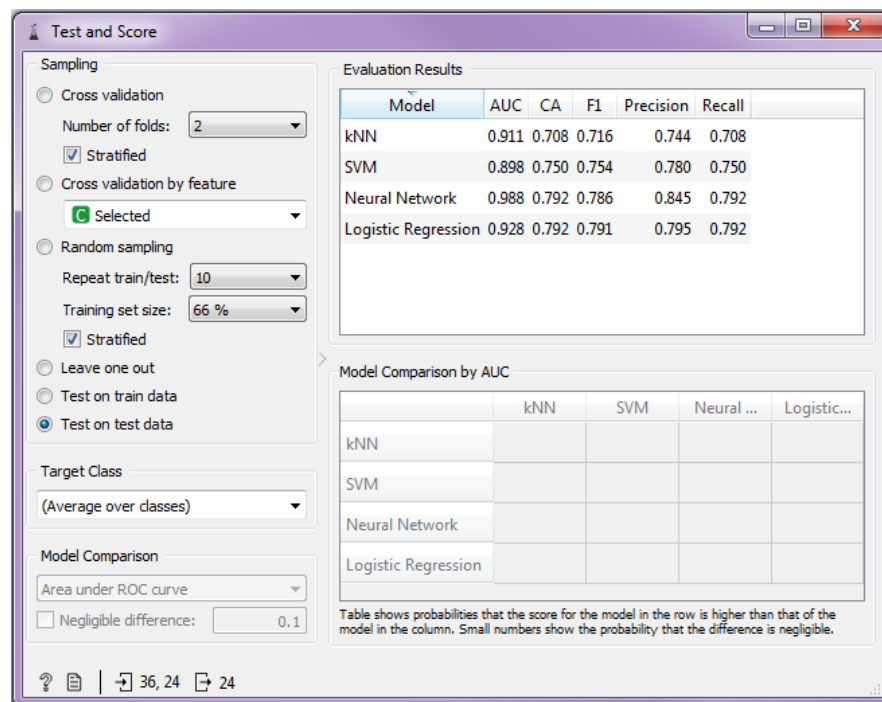


Figure 20 Test results for four classifiers

Step 9: We can save the developed model by adding **Save Model** widget and connect to our trained model. Then, double click the **Save Model** and click “Save as...” to save the model. For example, see below.

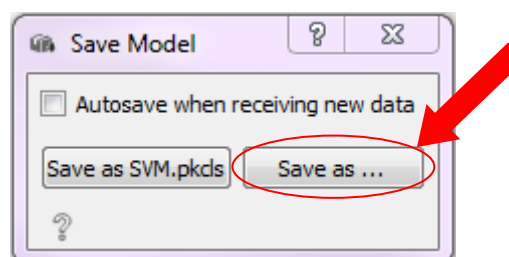
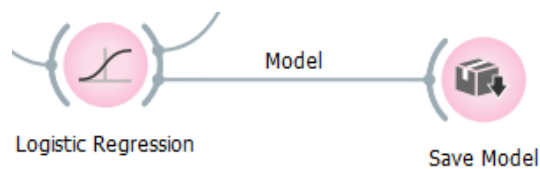


Figure 21 Save model workflow

PART C: Improving the trained model

Important Note: You are required to complete PART B before proceeding to Part C

Here for the reducing features purpose, we will remove some unnecessary features using several feature selection methods. Feature selection is an important step in understanding the best use of every feature.

Step 1: First, add a **Rank** and **Select Columns** widget. Then connect it together.

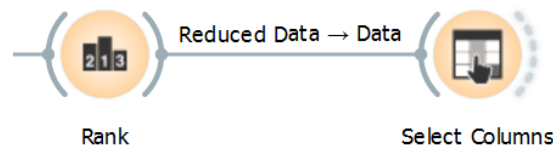


Figure 22 Rank to Select Columns connection

Step 2: Double click the **Rank** widget. There are many options of scoring methods to be select.

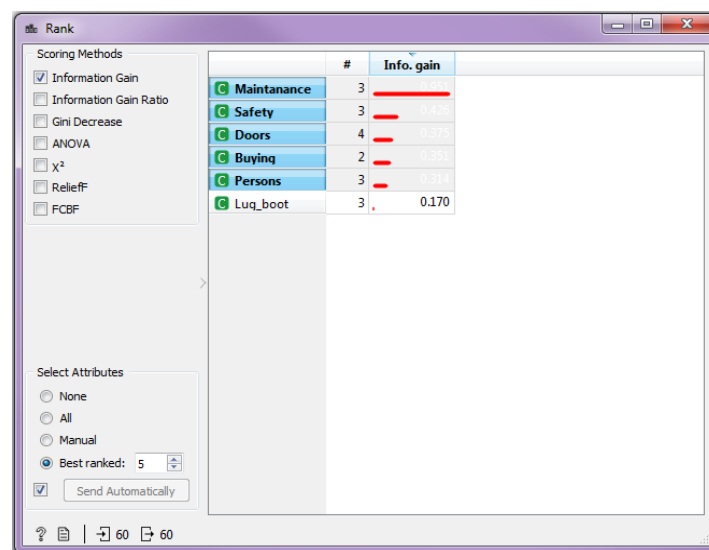


Figure 23 Rank widget

Here, we have selected the Information Gain for scoring method.

You can select from other scoring methods like:

- Information Gain Ratio
- Gini Decrease
- ANOVA
- Chi-squared test (χ^2)
- ReliefF
- FCBF

Step 3: Repeat all steps in PART B to train and test the model.

Basically after reducing the unnecessary features, the model developed will have a better accuracy, better results. Figure 24 and 25 proved that the models have better results after one (1) feature have been removed.

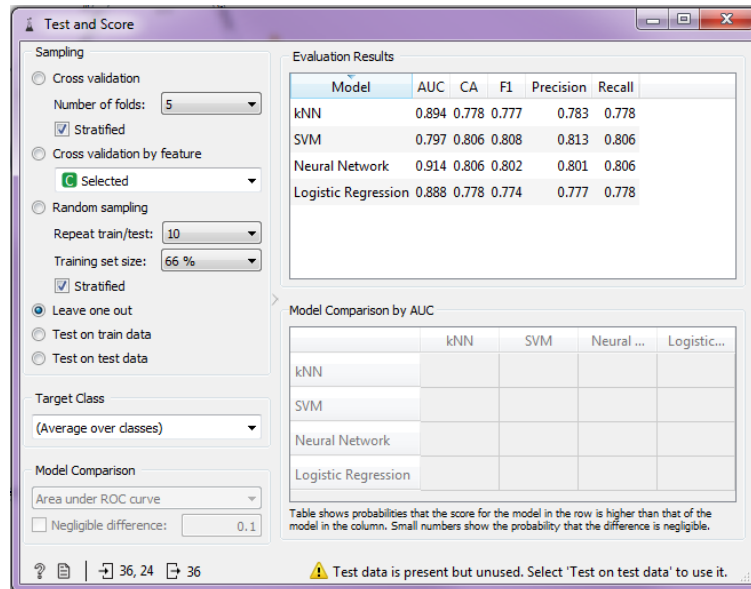


Figure 24 Train score for four classifiers with selected features

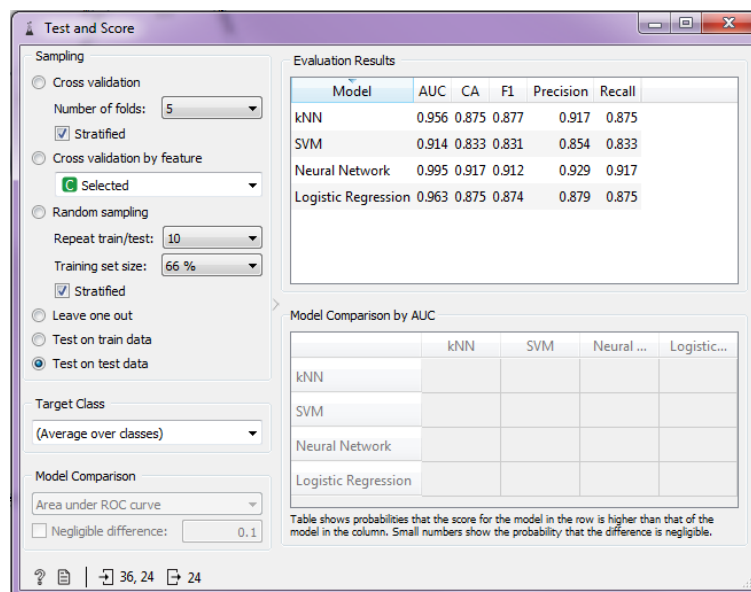
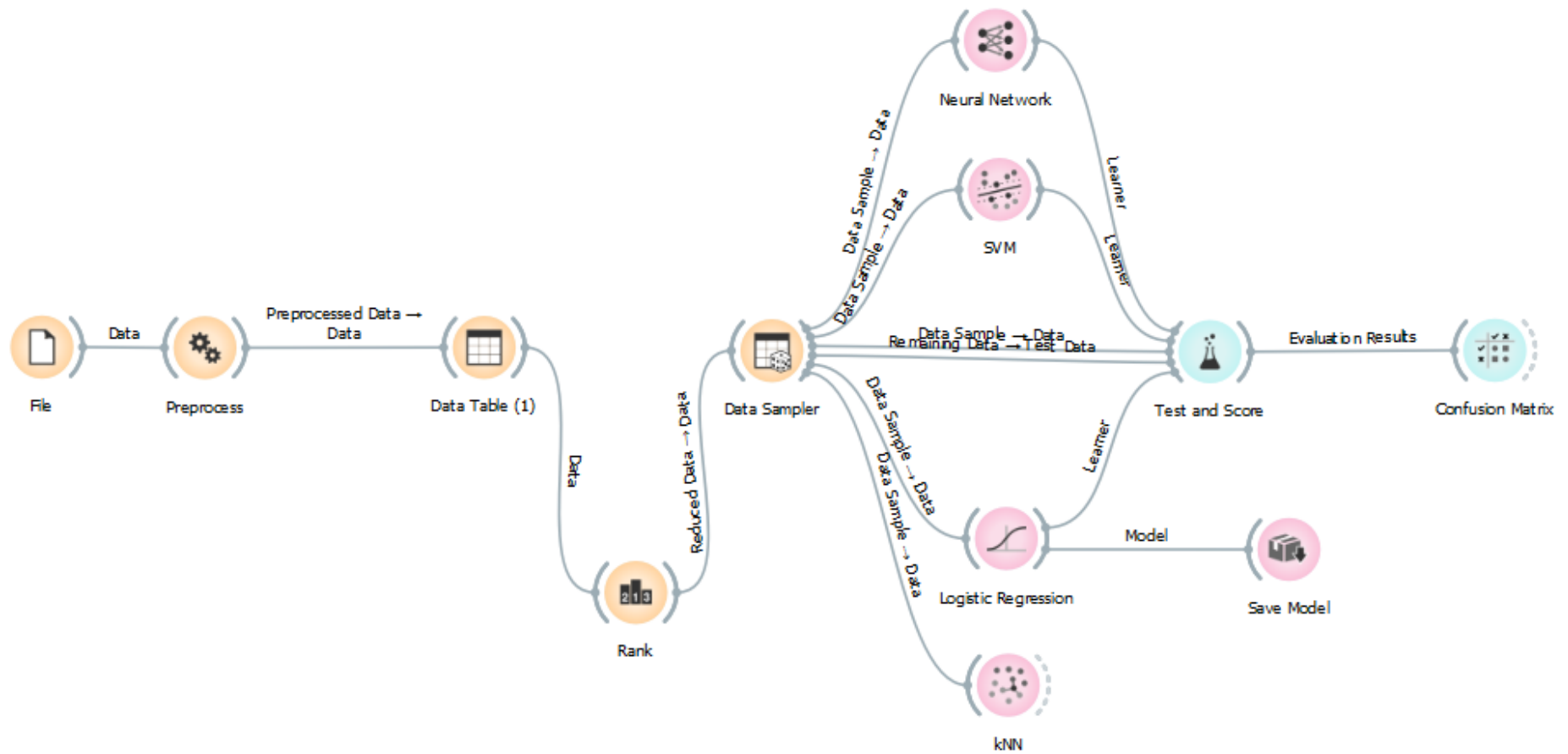


Figure 25 Test score for four classifiers with selected features

System workflow overview



Quick Exercise 1 (5 marks)

- 1. In which of the following scenarios would a high accuracy value suggest that the ML model is doing a good job?**
 - a) A deadly, but curable, medical condition afflicts .01% of the population. An ML model uses symptoms as features and predicts this affliction with an accuracy of 99.99%.
 - b) An expensive robotic chicken crosses a very busy road a thousand times per day. An ML model evaluates traffic patterns and predicts when this chicken can safely cross the street with an accuracy of 99.99%.
 - c) In the game of roulette, a ball is dropped on a spinning wheel and eventually lands in one of 38 slots. Using visual features (the spin of the ball, the position of the wheel when the ball was dropped, the height of the ball over the wheel), an ML model can predict the slot that the ball will land in with an accuracy of 4%.
- 2. Consider a classification model that separates email into two categories: "spam" or "not spam." If you raise the classification threshold, what will happen to precision?**
 - a) Probably increase.
 - b) Definitely decrease.
 - c) Definitely increase.
 - d) Probably decrease.

Question 2 (15 marks)

The previous tutorial evaluated the trained model against the training set, which does not provide a strong signal about the quality of your model. In this problem, you'll experiment with train sets, validation sets and test sets. Given dataset on COVID-19 cases from WHO database.

Learning objectives

- Split the dataset into 70% for train sets and 30% for test sets.
- Use multiple machine learning models to evaluate the datasets.
- Test the trained model with a test set to determine whether your trained model is overfitting.
- Try to reduce the features to obtain a better result and choose the best machine learning model.

REFERENCES

Medium, Introduction to machine learning
Analytics Vidhya, Building Machine Learning